



MYMOVE

* Park Smarter, Move Freely, Stay Private *

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1. Search & Reserve → 2. Real-Time IoT → 3. Stripe Checkout → 4. Privacy QR Chat

The Parking Bottleneck

GROWING CAR NUMBERS

Finding parking has become increasingly stressful as the number of vehicles continues to grow every day.

TRAFFIC CONGESTION

Areas such as shopping malls, office complexes, and university campuses often face ongoing parking challenges due to the large number of vehicles in these locations.

OUTDATED PARKING SYSTEM

Despite modern technology, many parking facilities still rely on manual checking, basic ticketing, and physical cards, making it difficult to provide real-time parking information.

Most popular car models and makes

Data as of 18 Aug 2026, 23:59

2026 ▼

#	CAR MODEL	TOTAL	#	CAR MAKE	TOTAL
1	Perodua Bezza	57,284	1	Perodua	190,137
2	Proton Saga	52,048	2	Proton	115,028
3	Perodua Axia	44,064	3	Toyota	62,413
4	Perodua Myvi	34,584	4	Honda	32,918
5	Perodua Alza	23,737	5	Chery	18,562



The Broken Parking Cycle

01

Outdated search systems

Outdated systems do not provide real-time availability or clear guidance, forcing drivers to continuously circle premises in search of unoccupied bays. (Kalašová et al., 2021)

02

Cruising congestion

Persistent double-parking disputes in Malaysia leave blocked vehicle owners with no fast or secure way to contact drivers who park behind them. (Haslinda et al., 2016)

03

Double-parking conflicts

DBKL received 1,009 public complaints and issued thousands of compound notices between January and May 2025 — enforcement alone cannot solve haphazard double-parking. (Anis, 2025)

● 1,009 DBKL complaints

04

Windshield privacy vulnerability

Leaving handwritten telephone numbers on vehicle windshields exposes drivers to harassment, identity theft, and unsolicited spam.



The Roadmap



AIM OF PROJECT

To develop a parking system that provides real-time information, finds parking spaces easily, reserves a spot, and communicates during double-parking situations to reduce stress, search time, and congestion.

RO1

Study & specify

Study existing parking systems, identify current issues faced by Malaysian drivers, analyze the limitations of current car parking applications, and define user requirement specifications and features through literature review and survey.

RO3

Evaluate & test

Evaluate the application usability and user satisfaction by conducting User Acceptance Testing (UAT) with real users and collecting feedback for further improvement.

RO2

Design & develop

Develop the myMove application (Flutter mobile app and React web admin) by implementing its core features, including real-time parking search, hardware-integrated spot monitoring, and booking.

The Ecosystem



Target audience

- Malaysian drivers and daily urban commuters
- Parking lot operators
- Facility security personnel



Mobile client

FLUTTER · DART

- Android application built with Flutter SDK and Dart
- Verified on physical Android devices (API level 21+)



Web frontends

REACT · VANILLA JS

- Web-guest scanner: zero-install browser portal for non-app users
- Web admin portal: facility dashboard built with React, TypeScript, and Vite



IoT edge

ESP32

- ESP32 microcontroller with HC-SR04 ultrasonic distance sensors mounted in custom 3D-printed brackets
- Live bay occupancy detection with sub-1.5s cloud synchronization



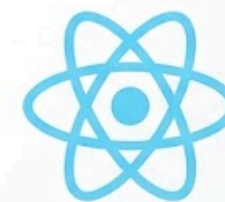
Payment integration

STRIPE

- Sandbox Stripe PaymentSheet
- Secure credit and debit card checkouts, and parking extensions



Flutter



React.js



ESP32
MICROCONTROLLER

stripe

The Missing Pieces

Application

Touch 'n Go eWallet

JomParking

Flexi Parking

myMove (proposed)

Core functionality

LPR parking, street parking, and travel passes.

Street council payment, map drag, compound billing.

On-street GPS payment, expiry alerts, 10 vehicles per profile.

IoT spot detection, spot booking, QR privacy chat, SOS alert.

System strengths

Convenient, reduces physical cash and cards.

Remote parking payment and time extensions.

Supports 46 municipal councils, prevents non-chargeable hour charges.

Sub-1.5s live sync, atomic Stripe reservation, zero-install web guest scanner, admin hub.

Identified weaknesses

Lacks card and eWallet integration, limited locations, no live slot availability.

Limited locations, online banking only, lacks off-street building integration.

Buggy navigation, registration errors, GPS inaccuracies, zero double-parking help.

Prototype ultrasonic sensors tested in a controlled diorama setup.

Proposed myMove improvements

Expands bay-level availability tracking and multi-location reservations.

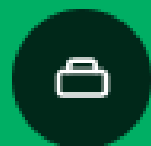
Introduces real-time bay selection (A1, A2) and hardware sensor sync.

Implements zero-install QR vehicle contact and real-time anonymous driver chat.

Delivers an end-to-end smart parking and privacy communication ecosystem.

What the Data Says

N = 37 participants | Age 18-45+ | University students & working commuters



DOUBLE-PARKING

ENCOUNTERED DOUBLE-PARKING
BLOCKAGES REGULARLY

89.2%



SEARCH DELAYS

STRUGGLED TO FIND AVAILABLE
SPOTS IN CROWDED AREAS

56.8%



BAY INFORMATION

DEMANDED BUILDING, FLOOR LEVEL,
AND LOT NUMBER DETAIL

73.0%



DIGITAL PAYMENT

REQUESTED SEAMLESS IN-APP
PAYMENT INTEGRATION

62.2%



QR-CODE CONTACT

VOTED FOR A DEDICATED QR CONTACT
SYSTEM FOR BLOCKED DRIVERS

51.4%

From Plan to Product

01

Requirements phase FYP1

Gathered parking problem data, analyzed competitor limitations, and formulated functional and non-functional specifications.

02

Design phase FYP1 / FYP2

Formulated UML use cases (Figure 4.1), 4-tier BaaS architecture, UI wireframes, and Firestore database schema models.

03

Development phase FYP2

Built the Flutter mobile app, ESP32 Arduino C++ firmware, React web admin portal, and web guest scanner.

04

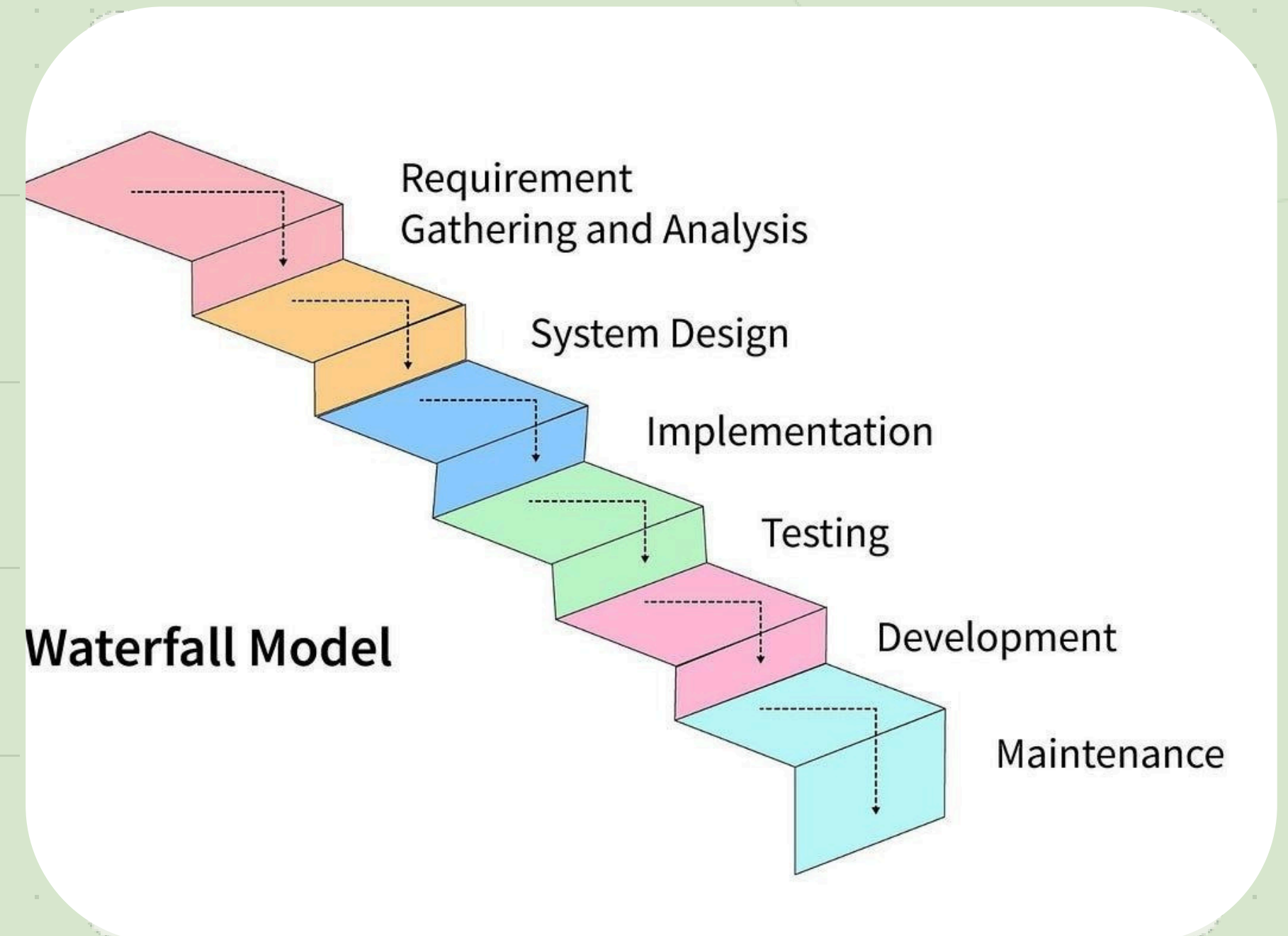
Testing phase FYP2

Executed unit testing (97.8% pass rate), integration testing (100% pass rate), latency benchmarking, and UAT with 15 users.

05

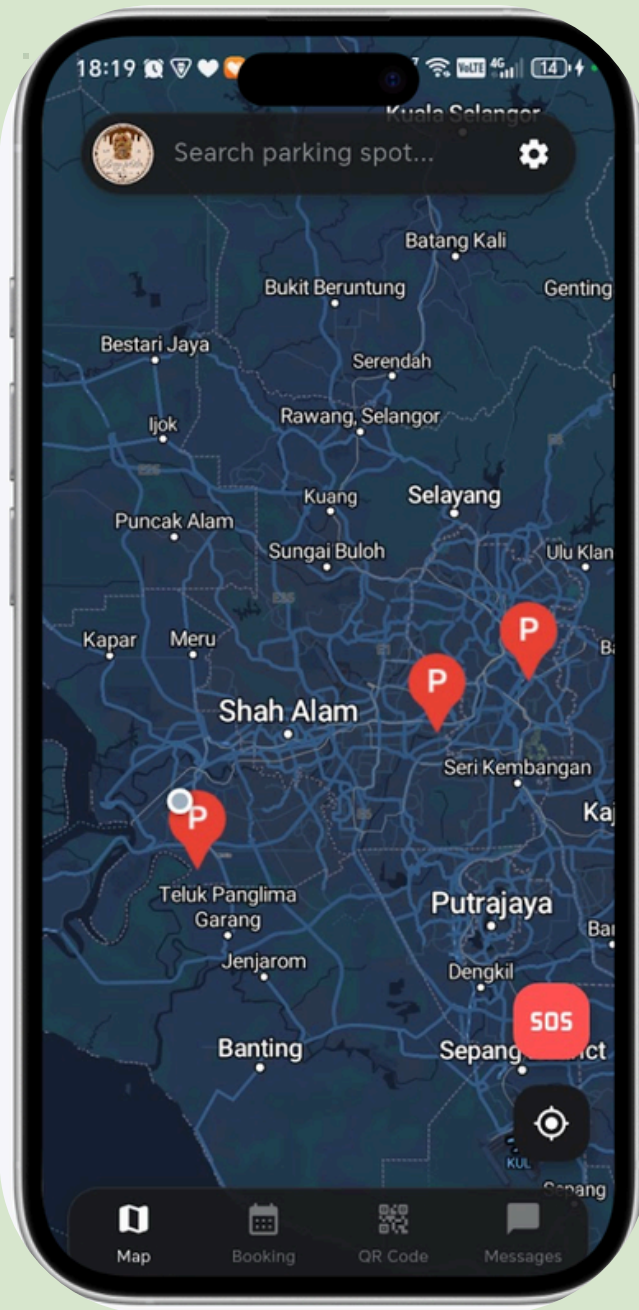
Deployment & maintenance FYP2

Cloud deployment to Firebase Hosting, Firestore security rules audit, and 3D diorama prototype fabrication.

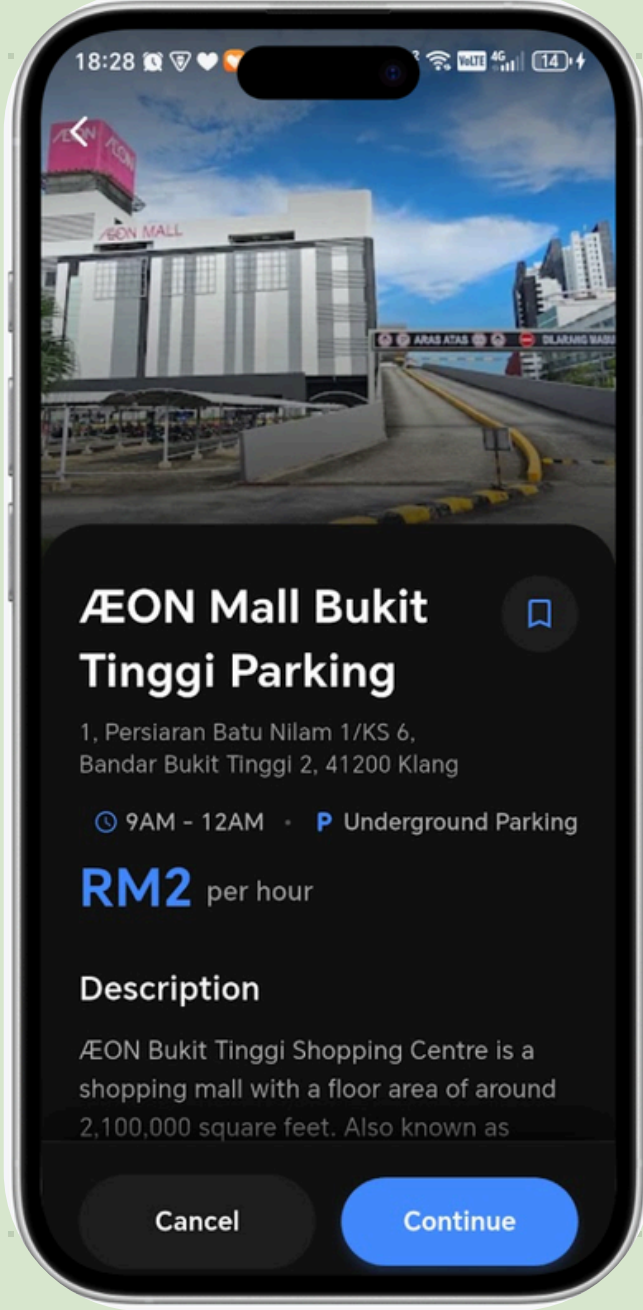


Tap, Book, Park

Smart tools. Seamless reservations. Total privacy.



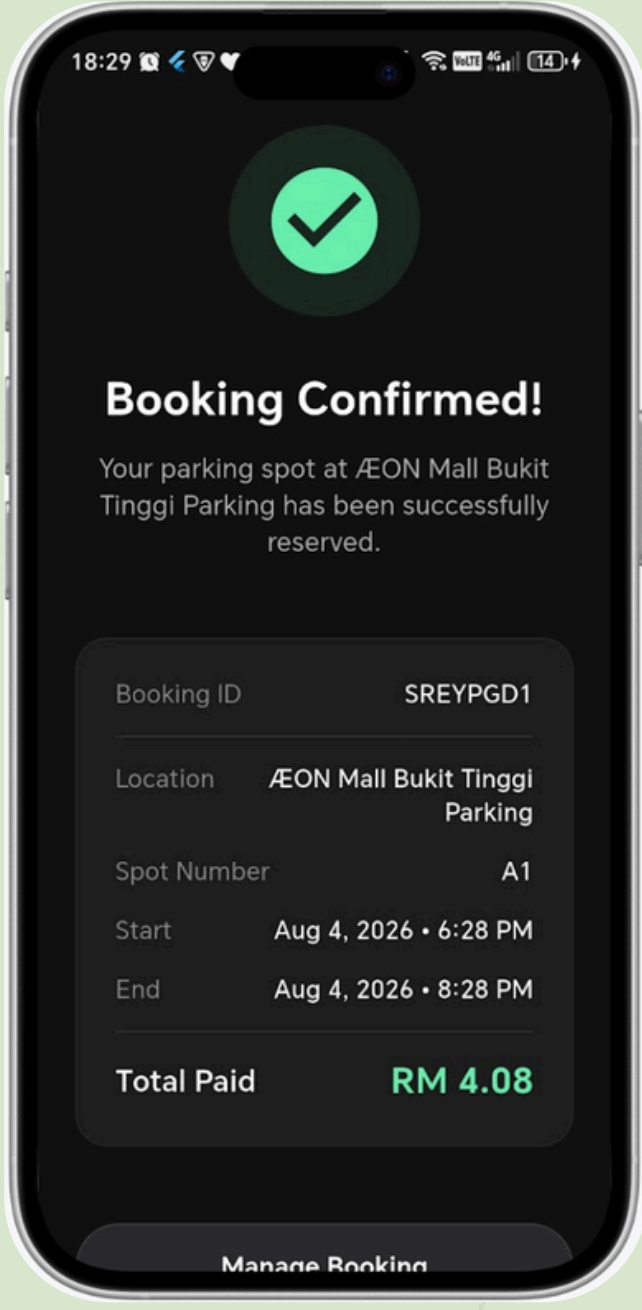
HOME & MAP



PARKING DETAILS



LIVE SPOT GRID



ACTIVE BOOKING



QR CODE SCAN

Managing the Floor, Answering SOS

Centralized facility management, real-time occupancy telemetry, and emergency dispatch.

FACILITY & SPOT MANAGEMENT

Configure parking facilities, define parking zones, set dynamic hourly tariffs, and override spot states manually.

REAL-TIME TELEMETRY & REVENUE

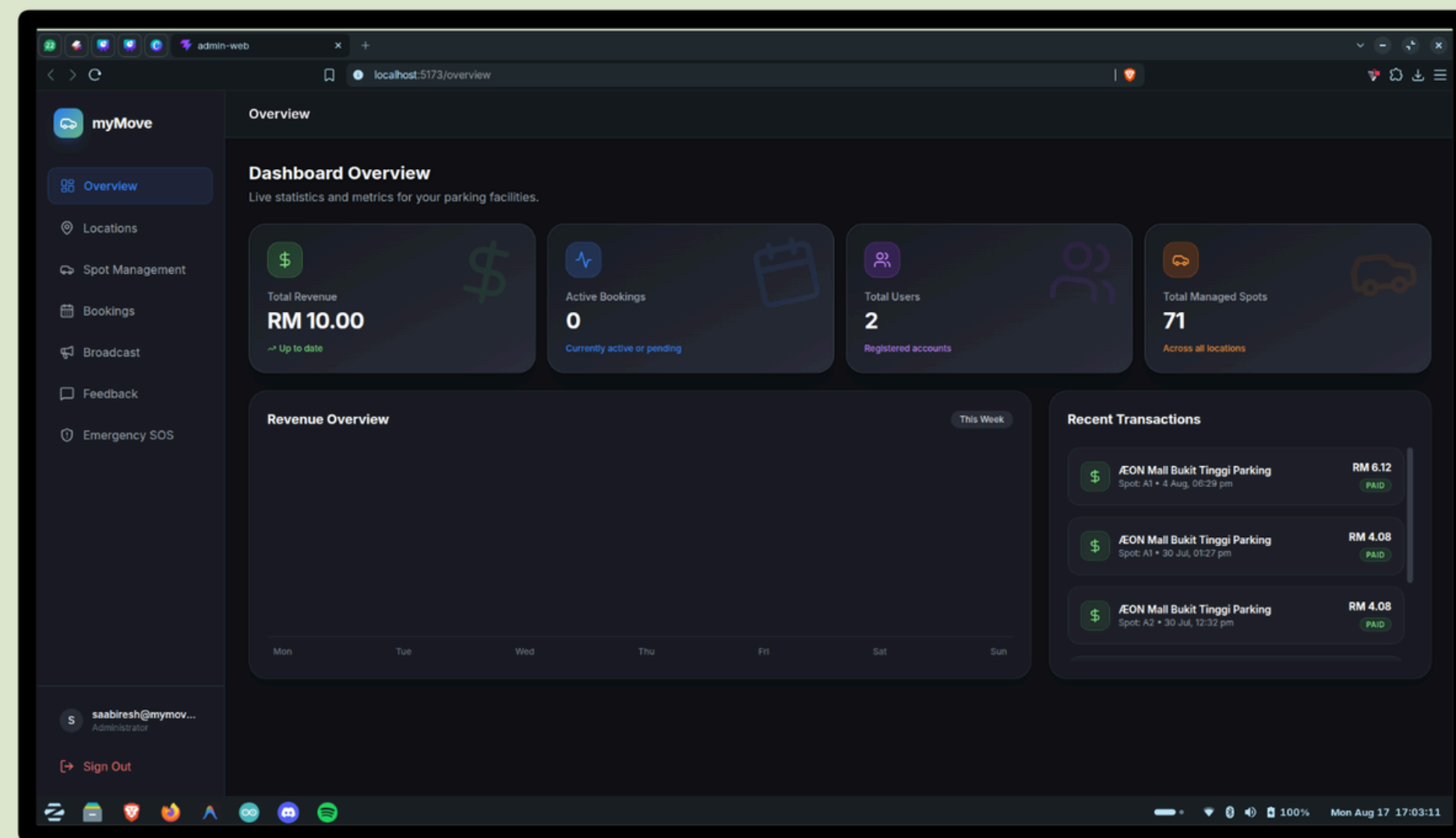
Live charts showing hourly occupancy peaks, weekly gross revenue, and total customer reservation volume.

EMERGENCY SOS CONSOLE

Dedicated visual and auditory panic hub alerting on-site security personnel instantly with driver GPS coordinates.

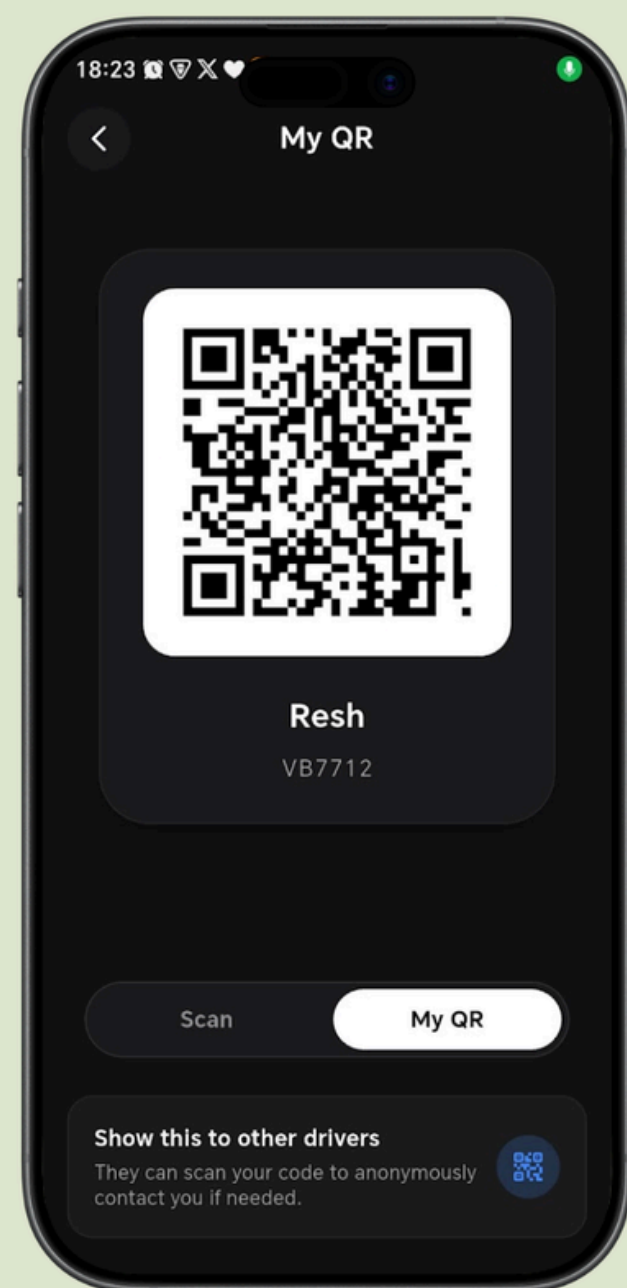
SYSTEM-WIDE PUSH BROADCASTS

Instantly dispatch facility announcements, maintenance closures, or weather warnings to all mobile app users.

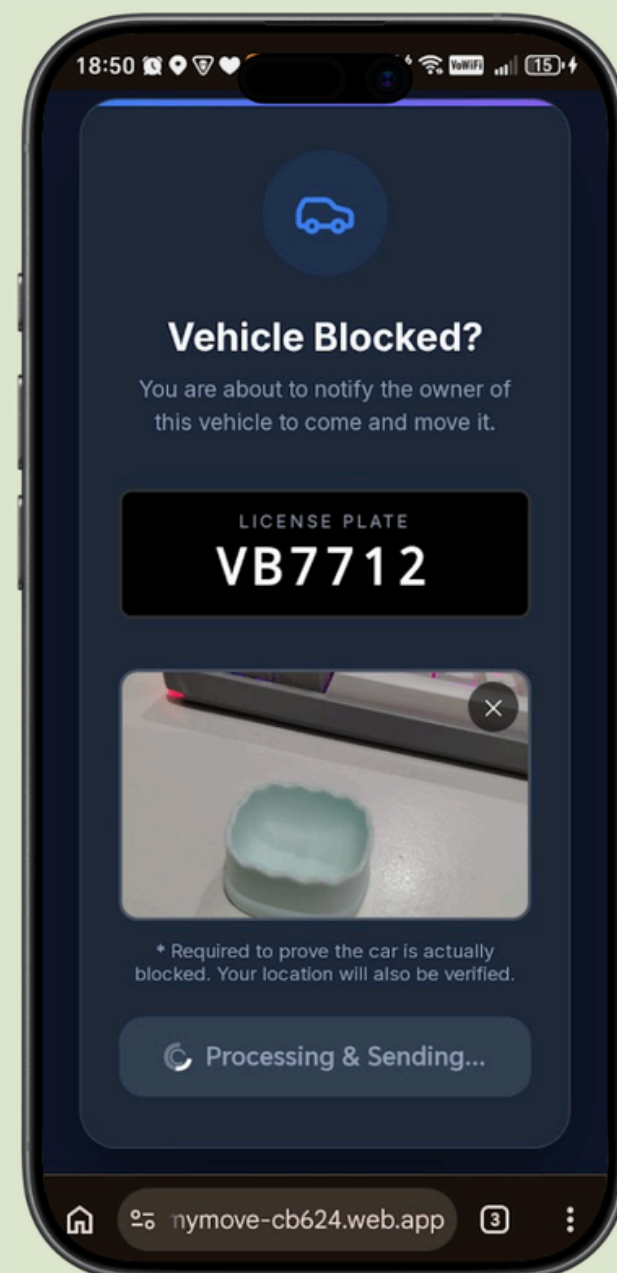


Privacy on the Windshield

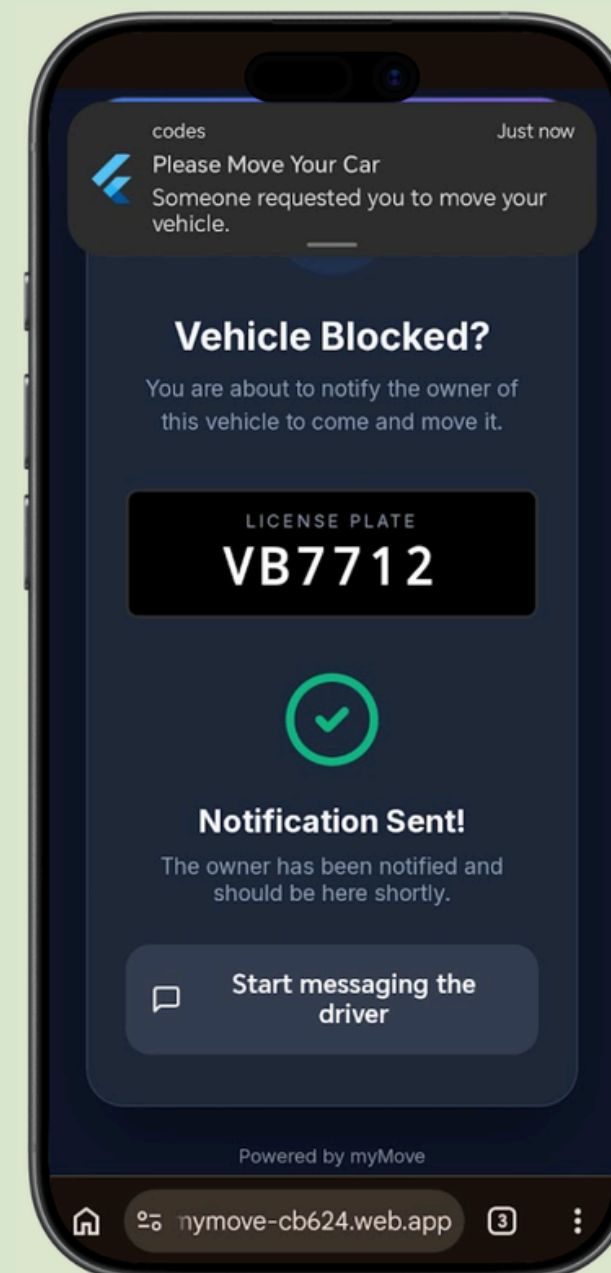
Zero app installation for guest drivers, and 100% privacy protection for registered vehicle owners.



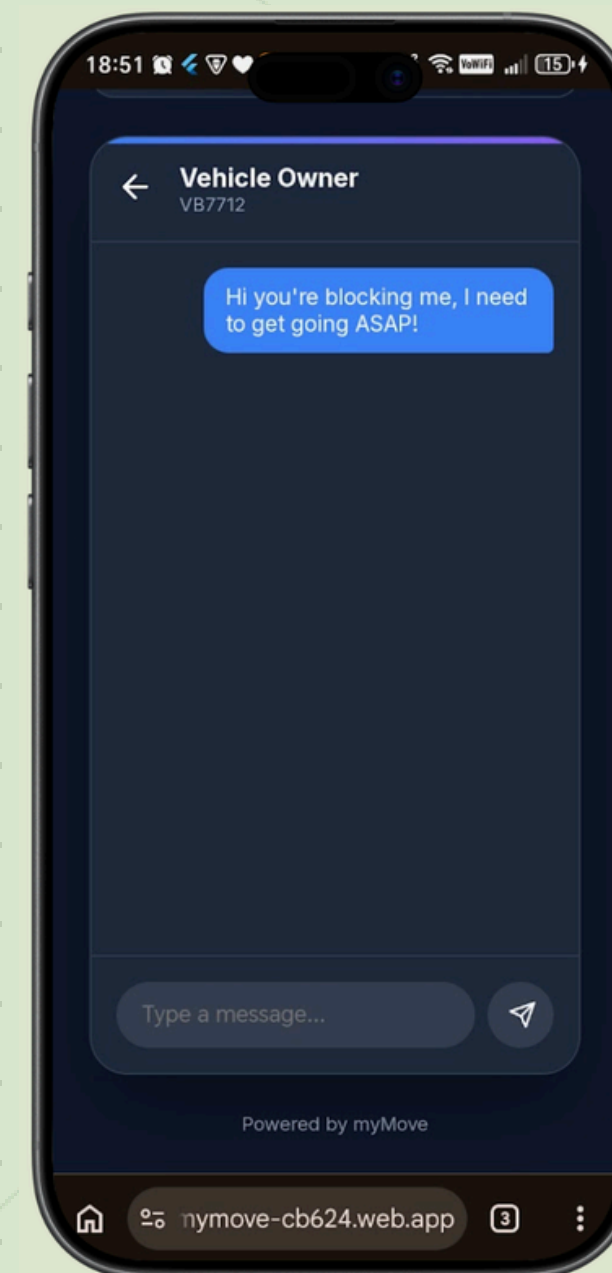
SCAN QR



UPLOAD PHOTO PROOF



ALERTED OWNER IN-APP



REAL-TIME WEB CHAT

Putting myMove to the Test



97.8%

UNIT TEST PASS RATE



1.1s

CLOUD SYNC LATENCY



4.81 / 5.00

UAT SATISFACTION SCORE

TESTING SUITE & LATENCY BENCHMARKS

Unit testing

Authentication, booking algorithms, GPS geofencing · Section 5.2

97.8%

Integration testing

Hardware ↔ Cloud ↔ Mobile ↔ Web tiers · Section 5.3

100%

Complete reservation & payment

100% transaction success rate

4.2s

IoT spot occupancy cloud sync

Sub-1.5s requirement achieved

1.1s

Double-parking QR notification

1.9s

SOS emergency broadcast speed

0.8s

USER ACCEPTANCE TESTING (N = 15)

Q4 Real-time grid accurately reflects availability 5.00 / 5

Q5 Stripe card reservation feels secure & convenient 5.00 / 5

Q7 System protects privacy by eliminating phone numbers 5.00 / 5

Q8 Overall, myMove is a highly effective solution 4.87 / 5

OVERALL MEAN USABILITY SCORE

4.81 / 5.00

CONCLUSION



Wrapping Up & Leveling Up

✓ RO1 · Study & requirements Achieved ✓ RO2 · System development Achieved ✓ RO3 · Evaluation & testing Achieved

SYSTEM LIMITATIONS

- 01 Continuous internet dependency**
Requires active data connection, with potential delays in deep underground basements.
- 02 Prototype sensor hardware**
HC-SR04 ultrasonic sensors tested in a tabletop diorama, sensitive to outdoor rain and dust.
- 03 Single-language scope**
System currently available exclusively in English.

FUTURE RECOMMENDATIONS & ROADMAP



AI license plate recognition

Overhead AI camera integration to eliminate individual ultrasonic sensor hardware.



Local payment gateways

Adding Touch 'n Go eWallet, FPX online banking, and DuitNow QR.



AR indoor navigation

AR camera guidance directing drivers to their exact reserved bay.



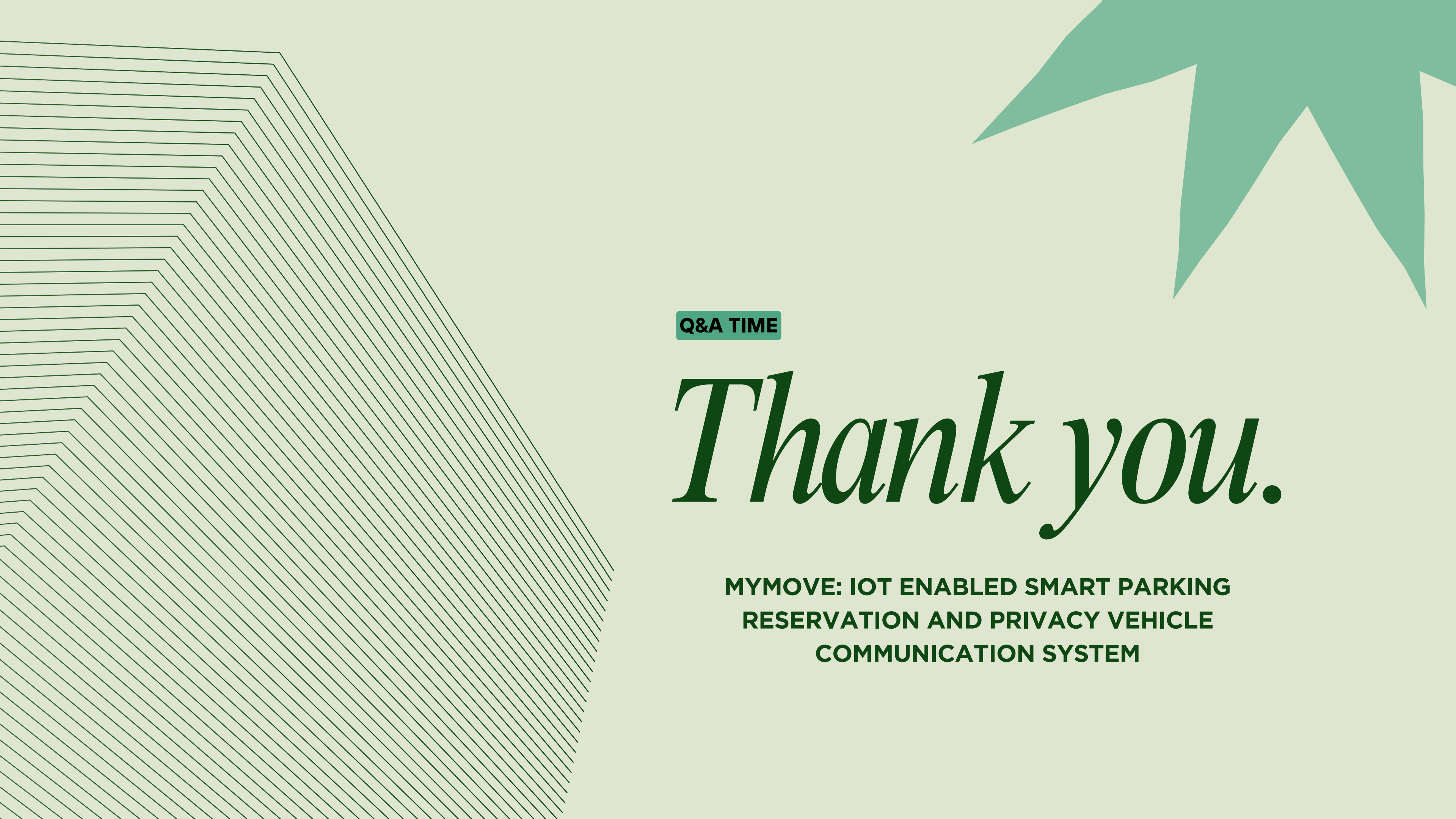
Multi-language support

Adding Bahasa Melayu and Mandarin localization for wider Malaysian adoption.

Thank you

Questions & discussion welcome

Q&A



Q&A TIME

Thank you.

**MYMOVE: IOT ENABLED SMART PARKING
RESERVATION AND PRIVACY VEHICLE
COMMUNICATION SYSTEM**